

Notes: Beaudry and Portier (AER, 2006)
Stock Prices, News, and Economic Fluctuations

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1 Big picture

This paper asks whether stock prices contain useful information about future productivity growth and whether this information helps explain business cycle fluctuations. The central idea is that stock prices are forward-looking. If they jump today without any contemporaneous measured TFP innovation, and if that same movement later predicts permanent TFP growth, then stock prices may be revealing **news** about future technological opportunities.

The paper's key empirical fact is striking: two different shocks, identified in two different ways, look almost the same. The first shock is an innovation in stock prices that is orthogonal to current TFP. The second shock is the innovation that drives long-run TFP. In the data, these two shocks are nearly colinear and generate nearly identical impulse responses. This is hard to reconcile with a simple model in which surprise technology shocks immediately move TFP and other demand or monetary shocks move stock prices. It is much more consistent with a model in which agents receive news about future productivity before measured productivity rises.

The economic interpretation is an **expectation-driven boom**: agents learn about future improvements in technology, stock prices rise immediately, and consumption, investment, and hours respond before the measured productivity improvement is fully realized.

2 Incremental contribution of the paper

The incremental contribution is not simply to show that stock prices forecast real activity. That fact was already known from earlier work on stock returns and future production growth. The contribution is to use stock prices and TFP together in a VAR framework to distinguish between surprise productivity shocks and news about future productivity.

The paper contributes in five main ways.

1. It introduces a simple sequential use of short-run and long-run restrictions. Rather than imposing both simultaneously, the paper separately recovers a stock-price-news shock and a long-run TFP shock, then compares them.
2. It shows that a stock-price innovation with no contemporaneous effect on TFP is almost identical to the shock that explains permanent movements in TFP.
3. It provides a theoretical interpretation based on delayed diffusion of technological innovations: stock prices react when the information arrives, while measured TFP responds later.
4. It shows that the result is robust to alternative TFP measures, including measures adjusted for variable factor utilization.
5. It extends the analysis beyond the bivariate TFP-stock-price system to systems that include consumption, investment, output, and hours, showing that the identified news shock explains standard business-cycle comovement.

The paper therefore pushes macroeconomic VAR analysis toward a news-shock view of fluctuations. Business cycles may be driven by information about future fundamentals, not only by contemporaneous changes in fundamentals.

3 Data and measurement

The main sample uses postwar U.S. data from 1948:Q1 to 2000:Q4. The two core variables in the baseline bivariate system are:

- **Stock prices:** the quarterly S&P 500 Composite Stock Price Index, deflated by the implicit price deflator for nonfarm private business GDP and converted into per-capita terms.
- **Measured TFP:** a nonfarm private business total factor productivity measure constructed from output, hours, labor share, and capital services.

The paper then considers alternative TFP measures to address the concern that measured TFP can move because of factor utilization rather than true technological opportunities. The important alternatives are:

- the annual Basu-Fernald-Kimball measure, which adjusts for utilization and other biases;
- an adjusted quarterly TFP series constructed using the BLS measure of capacity utilization.

The higher-dimensional systems add real macro variables: consumption, investment, output, and hours worked. Consumption is real personal consumption of nondurables and services per capita. Investment is durable consumption plus fixed private domestic investment per capita.

4 Models and empirical specifications

4.1 Reduced-form system

The starting point is a reduced-form moving-average representation for the bivariate system:

$$\begin{pmatrix} \Delta TFP_t \\ \Delta SP_t \end{pmatrix} = C(L) \begin{pmatrix} \mu_{1t} \\ \mu_{2t} \end{pmatrix}, \quad (1)$$

where SP_t is the stock price index and TFP_t is measured total factor productivity. In practice, the moving-average representation is obtained from a vector error correction model because stock prices and TFP are treated as likely cointegrated integrated processes.

4.2 Short-run orthogonalization

The first orthogonalized representation is

$$\begin{pmatrix} \Delta TFP_t \\ \Delta SP_t \end{pmatrix} = \Gamma(L) \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix}. \quad (2)$$

The identifying restriction is that the second shock, ε_{2t} , has no contemporaneous impact on TFP. Thus, ε_{2t} is interpreted as an innovation in stock prices that is orthogonal to current measured productivity. This is the paper's empirical proxy for news that affects expectations before it affects measured TFP.

4.3 Long-run orthogonalization

The second orthogonalized representation is

$$\begin{pmatrix} \Delta TFP_t \\ \Delta SP_t \end{pmatrix} = \tilde{\Gamma}(L) \begin{pmatrix} \tilde{\varepsilon}_{1t} \\ \tilde{\varepsilon}_{2t} \end{pmatrix}. \quad (3)$$

Here the long-run restriction is that $\tilde{\varepsilon}_{2t}$ has no long-run effect on TFP. Therefore, $\tilde{\varepsilon}_{1t}$ is the shock that drives permanent movements in TFP.

The core empirical comparison is between ε_{2t} and $\tilde{\varepsilon}_{1t}$. If stock-price innovations that are unrelated to current TFP are actually news about future TFP, these two series should be highly correlated and should generate similar impulse responses.

4.4 Delayed-diffusion interpretation

The paper presents a simple model in which measured TFP is the sum of a permanent component and a transitory component:

$$\theta_t = D_t + v_t, \quad (4)$$

with

$$D_t = \sum_{i=0}^{\infty} d_i \eta_{1,t-i}, \quad d_0 = 0, \quad d_i \leq d_{i+1}, \quad \lim_{i \rightarrow \infty} d_i = 1, \quad (5)$$

and

$$v_t = \rho v_{t-1} + \eta_{2t}. \quad (6)$$

The innovation η_{1t} is news about a permanent technological improvement, but the improvement diffuses only gradually into measured productivity. The assumption $d_0 = 0$ means that the news has no immediate impact on measured TFP. Stock prices, however, can move immediately because they are forward-looking.

5 Identification and empirical design

The paper's empirical design is descriptive but disciplined by model implications. It does not claim that a single orthogonalization is a structural model. Instead, it asks whether two very different identification schemes recover the same underlying shock.

1. **Recover a short-run stock-price innovation.** Estimate a VECM for TFP and stock prices and identify ε_{2t} as the shock that moves stock prices but not current TFP.
2. **Recover a long-run TFP shock.** Estimate the same system and identify $\tilde{\varepsilon}_{1t}$ as the shock that has a permanent effect on TFP.
3. **Compare the shocks.** If ε_{2t} and $\tilde{\varepsilon}_{1t}$ are nearly identical, the short-run stock-price innovation is revealing information about long-run productivity.

4. **Inspect impulse responses.** A news interpretation predicts that stock prices jump immediately, while TFP may respond with a delay, especially when TFP is adjusted for variable factor utilization.
5. **Extend the system.** Add consumption, investment, output, and hours to ask whether the same shock produces standard business-cycle comovement.

A useful way to summarize the identifying logic is this: if a shock is orthogonal to current TFP but eventually explains permanent TFP movements, then it is difficult to treat it as a standard surprise TFP shock. It looks instead like news about future TFP.

6 Main results

6.1 The two identified shocks are almost the same

In the baseline bivariate system, the shock identified as an innovation in stock prices with no contemporaneous TFP effect and the shock identified as the long-run TFP shock are almost perfectly colinear. The paper reports a correlation around 0.97 between these two shock series. Their impulse responses for TFP and stock prices are also nearly identical.

6.2 Stock prices anticipate future TFP

The baseline impulse responses show that stock prices react immediately and strongly. Measured TFP then increases gradually and permanently. This pattern suggests that the stock market prices in future productivity growth before the productivity increase fully appears in measured TFP.

6.3 Utilization-adjusted TFP strengthens the news interpretation

When the paper uses the annual Basu-Fernald-Kimball utilization-adjusted TFP series, the same two shocks remain almost identical: their correlation is about 0.989. But the timing of TFP is more revealing. TFP does not increase immediately; the point estimates suggest that TFP begins to rise several years after the stock market response. This is exactly the pattern predicted by a delayed-diffusion news model.

The BLS capacity-utilization adjustment produces similar qualitative evidence. Adjustment changes the timing of measured TFP: unadjusted TFP appears to respond faster, while adjusted TFP responds only after a longer delay. This suggests that some short-run measured TFP response in standard data reflects variable factor utilization rather than a true immediate technology improvement.

6.4 The shock generates business-cycle comovement

When consumption, investment, output, and hours are added to the system, the same news shock produces a temporary boom. Stock prices, consumption, and hours respond positively before the full TFP improvement is observed. This pattern fits the idea that agents react to expected future productivity by increasing demand and activity today.

6.5 The shock is quantitatively important

The variance decompositions show that the identified news shock accounts for a substantial fraction of forecast-error variance in consumption, investment, output, and hours. The paper's conclusion is therefore not merely about timing; the shock appears quantitatively important for business-cycle fluctuations.

7 Figures

7.1 Baseline bivariate evidence

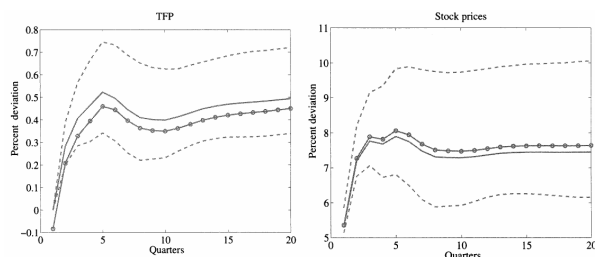


FIGURE 1. IMPULSE RESPONSES TO SHOCKS ε_2 AND $\tilde{\varepsilon}_1$ IN THE (TFP, SP) VECM

Notes: In both panels of this figure, the bold line represents the point estimate of the responses to a unit ε_2 shock (the shock that does not have instantaneous impact of TFP in the short-run identification). The line with circles represents the point estimate of the responses to a unit $\tilde{\varepsilon}_1$ shock (the shock that has a permanent impact on TFP in the long-run identification). Both identifications are done in the baseline bivariate specification (five lags and one cointegrating relation). The unit of the vertical axis is percentage deviation from the situation without shock. Dotted lines represent the 10-percent and 90-percent quantiles of the distribution of the impulse response functions (IRFs) in the case of the short-run identification, this distribution being the Bayesian simulated distribution obtained by Monte-Carlo integration with 2,500 replications, using the approach for just-identified systems discussed in Thomas J. Doan (1992).

(a) Figure 1. Baseline impulse responses in the (TFP, SP) VECM

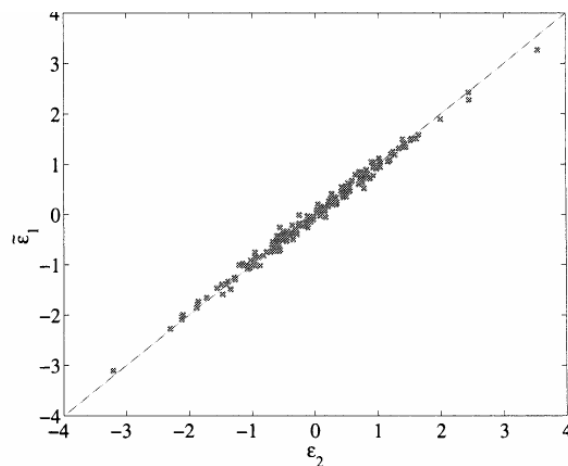


FIGURE 2. PLOT OF ε_2 AGAINST $\tilde{\varepsilon}_1$ IN THE (TFP, SP) VECM

Notes: This figure plots ε_2 against $\tilde{\varepsilon}_1$. Both shocks are obtained from the baseline bivariate specification (five lags and one cointegrating relation). The straight line is the 45-degree line.

(b) Figure 2. Baseline scatter of ε_2 and $\tilde{\varepsilon}_1$

Read:

- Figure 1 shows that the stock-price innovation with no contemporaneous TFP effect and the long-run TFP shock generate nearly identical responses of TFP and stock prices.
- Stock prices jump immediately; TFP rises gradually and persistently.
- Figure 2 shows the two shock series lying very close to the 45-degree line.

Economic message: A shock that is orthogonal to current TFP nevertheless looks like the shock that drives long-run TFP. This is the paper's central evidence for news about future productivity.

7.2 Utilization-adjusted annual TFP

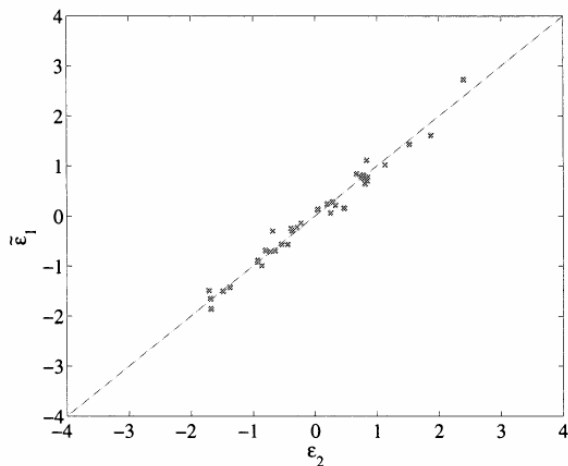


FIGURE 3. PLOT OF ε_2 AGAINST $\tilde{\varepsilon}_1$ IN THE (TFP, SP) VECM, USING BASU ET AL. (2002) MEASURE OF TFP (ANNUAL, 1949–1989)

Notes: This figure plots ε_2 against $\tilde{\varepsilon}_1$. Both shocks are obtained from the baseline annual specification (two lags and one cointegrating relation). The straight line is the 45-degree line.

(a) Figure 3. Scatter using Basu et al. adjusted TFP

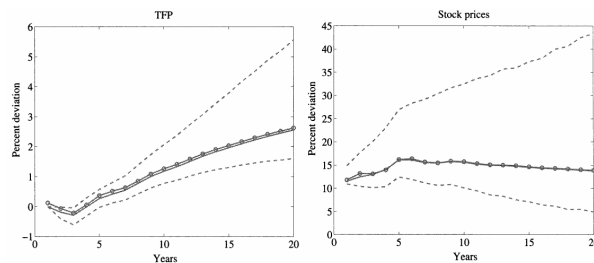


FIGURE 4. IMPULSE RESPONSES TO SHOCKS ε_2 AND $\tilde{\varepsilon}_1$ IN THE (TFP, SP) VECM, USING BASU ET AL. (2002) MEASURE OF TFP (ANNUAL, 1949–1989)

Notes: In both panels of this figure, the bold line represents the point estimate of the responses to a unit ε_2 shock (the shock that does not have instantaneous impact on TFP in the short-run identification). The line with circles represents the point estimate of the responses to a unit $\tilde{\varepsilon}_1$ shock (the shock that has a permanent impact on TFP in the long-run identification). Both identifications are done in the baseline annual specification (two lags and one cointegrating relation). The unit of the vertical axis is percentage deviation from the situation without shock. Dotted lines represent the 10-percent and 90-percent quantiles of the distribution of the IRF in the case of the short-run identification, this distribution being the Bayesian simulated distribution obtained by Monte-Carlo integration with 2,500 replications, using the approach for just-identified systems discussed in Doan (1992).

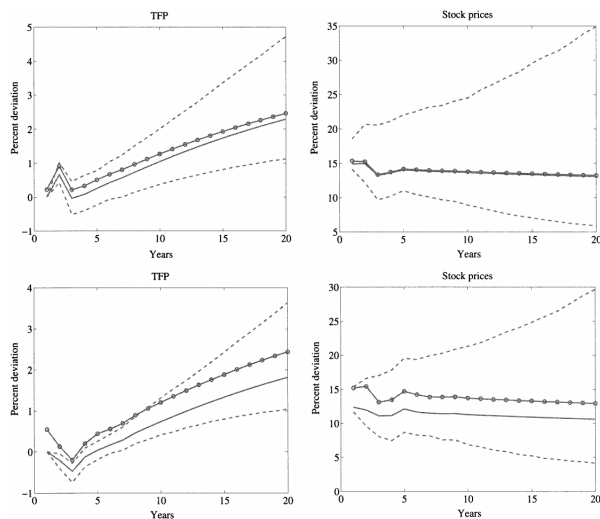
(b) Figure 4. Impulse responses using Basu et al. adjusted TFP

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- Figure 3 shows that the two identified shocks remain almost perfectly aligned using the Basu et al. adjusted TFP series.
- Figure 4 shows that stock prices respond immediately, while adjusted TFP rises only after a delay.
- The use of utilization-adjusted TFP reduces the concern that the result is driven by short-run changes in factor utilization.

Economic message: The adjusted TFP evidence is especially supportive of a news view: information arrives first, asset prices respond first, and measured productivity catches up later.

7.3 Annual adjusted versus unadjusted specifications



(a) Figure 5. Annual IRFs with and without TFP adjustment

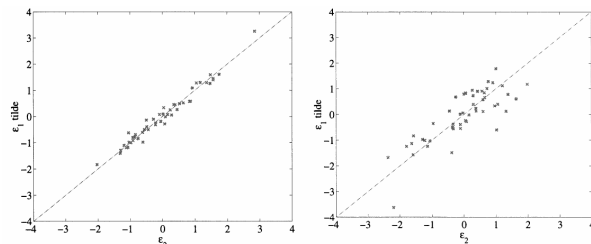


FIGURE 6. PLOT OF e_2 AGAINST e_1 IN THE (TFP, SP) VECM, USING ANNUAL OBSERVATIONS (1948–2000), WITHOUT ADJUSTING TFP FOR CAPACITY UTILIZATION (LEFT PANEL) OR WITH TFP ADJUSTMENT (RIGHT PANEL).
Notes: Each panel of this figure plots e_2 against e_1 . Both shocks are obtained from the baseline annual specification (two lags and one cointegrating relation), with either TFP or TFP^A . The straight line is the 45-degree line.

(b) Figure 6. Annual shock correlations with and without TFP adjustment

Read:

- Figure 5 compares annual impulse responses based on unadjusted TFP and capacity-utilization-adjusted TFP.
- Adjustment changes the timing of the TFP response: the adjusted measure displays a more delayed response.
- Figure 6 shows that the relationship between the two identified shocks remains strong, though somewhat weaker when TFP is adjusted.

Economic message: The exact timing of TFP depends on measurement, but the broader fact remains: stock-price news is closely related to future permanent productivity movements.

7.4 Quarterly adjusted TFP and three-variable systems

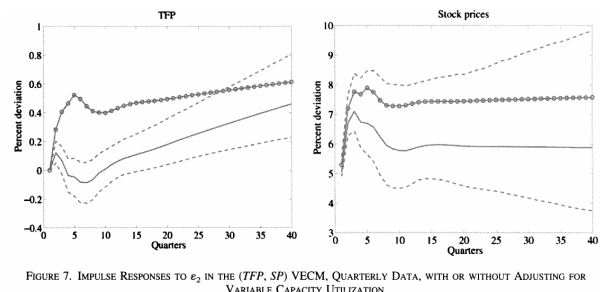


FIGURE 7. IMPULSE RESPONSES TO e_2 IN THE (TFP, SP) VECM, QUARTERLY DATA, WITH OR WITHOUT ADJUSTING FOR VARIABLE CAPACITY UTILIZATION
Notes: In both panels of this figure, the bold line represents the point estimate of the responses to a unit e_2 shock (the shock that does not have instantaneous impact on TFP in the short-run identification) in the VECM with adjusted TFP. The line with circles represents the point estimate of the responses to a unit e_2 shock in the VECM with nonadjusted TFP. The specification is the baseline bivariate one (five lags and one cointegrating relation). The unit of the vertical axis is percentage deviation from the situation without shock. Dotted lines represent the 10-percent and 90-percent quantiles of the distribution of the IRF in the VECM with adjusted TFP, this distribution being the Bayesian simulated distribution obtained by Monte-Carlo integration with 2,500 replications, using the approach for just-identified systems discussed in Doan (1992).

(a) Figure 7. Quarterly IRFs with adjusted and unadjusted TFP

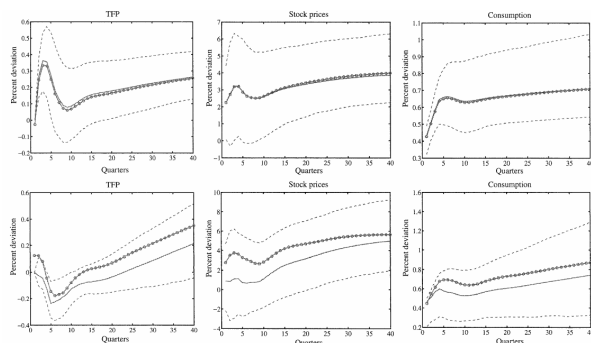


FIGURE 8. IMPULSE RESPONSES TO e_2 AND e_1 IN THE (TFP, SP, C) VECM, WITHOUT ADJUSTING TFP FOR CAPACITY UTILIZATION (UPPER PANELS) OR WITH TFP ADJUSTMENT (LOWER PANELS)

Notes: In each panel of this figure, the bold line represents the point estimate of the responses to a unit e_2 shock (the shock that does not have instantaneous impact on TFP in the short-run identification) in the VECM with adjusted TFP. The line with circles represents the point estimate of the responses to a unit e_2 shock in the VECM with nonadjusted TFP. The specification is the baseline bivariate one (five lags and one cointegrating relation). The unit of the vertical axis is percentage deviation from the situation without shock. Dotted lines represent the 10-percent and 90-percent quantiles of the distribution of the IRF in the VECM with adjusted TFP, this distribution being the Bayesian simulated distribution obtained by Monte-Carlo integration with 2,500 replications, using the approach for just-identified systems discussed in Doan (1992).

(b) Figure 8. IRFs in the (TFP, SP, C) system

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- Figure 7 shows the quarterly evidence: the stock price response is robust, while adjusted TFP responds more slowly than unadjusted TFP.
- Figure 8 adds consumption to the VAR and shows that consumption responds positively to the identified shock.
- The two orthogonalizations again deliver similar dynamics in the larger system.

Economic message: News shocks are not only asset-pricing events. They are associated with real macroeconomic comovement, especially in consumption.

7.5 Four-variable systems and variance decompositions

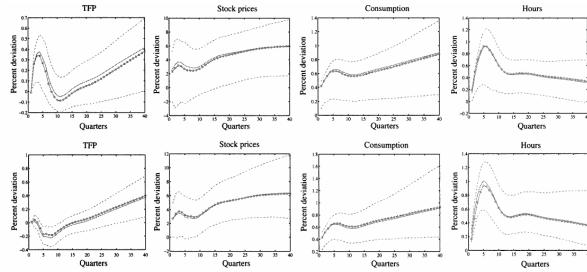


FIGURE 9. IMPULSE RESPONSES TO e_2 AND e_1 IN THE (TFP, SP, C, H) VECM, WITHOUT (UPPER PANELS) OR WITH (LOWER PANELS) ADJUSTING TFP FOR CAPACITY UTILIZATION

Notes: In each panel of this figure, the bold line represents the point estimate of the responses to a unit e_2 shock (the shock that does not have instantaneous impact on TFP in the short-run identification). The line with circles represents the point estimate of the responses to a unit e_1 shock (the shock that has a permanent impact on TFP in the long-run identification). In this system with hours, both identifications are done in a specification with five lags and three cointegrating relations, i.e., a VAR in levels. The unit of the vertical axis is percentage deviation from the situation without shock. Dotted lines represent the 10-percent and 90-percent quantiles of the distribution of the IRF in the case of the short-run identification, this distribution being the Bayesian simulated distribution obtained by Monte-Carlo integration with 2,500 replications, using the

(a) Figure 9. IRFs in the (TFP, SP, C, H) system

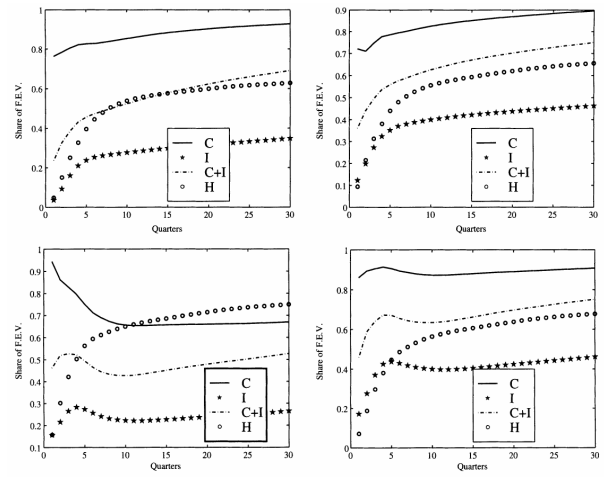


FIGURE 10. SHARE OF THE FORECAST ERROR VARIANCE (F.E.V.) OF CONSUMPTION (C), INVESTMENT I , OUTPUT ($C + I$), AND HOURS (H) ATTRIBUTABLE TO e_2 (LEFT PANELS) AND TO e_1 (RIGHT PANELS) IN VECMs, WITH NONADJUSTED TFP (TOP PANELS) OR ADJUSTED TFP (BOTTOM PANELS)

Notes: This figure has four panels. The left panels display the share of the forecast variance of consumption and investment that is attributable to e_2 (short-run identification) in the (TFP, SP, C, I) VECM (five lags and three cointegrating relations), of output ($C + I$) in the $(TFP, SP, C, C + I)$ VECM (five lags and three cointegrating relations), and of hours (H) in the $(TFP,$

(b) Figure 10. Forecast-error variance decomposition

Read:

- Figure 9 adds hours to the system and shows that hours rise after the news shock, creating a temporary boom before full productivity growth arrives.
- Consumption and stock prices respond strongly, while adjusted TFP again responds with delay.
- Figure 10 shows that the news shock explains a substantial share of forecast-error variance for consumption, investment, output, and hours.

Economic message: The news shock is not a small residual phenomenon. It appears to be an important driver of business-cycle dynamics and a source of the joint movement of asset prices and real activity.

8 Short summary

The paper’s main result is that stock-price innovations orthogonal to current TFP are nearly the same as long-run TFP shocks. This empirical equivalence is hard to reconcile with standard models in which productivity shocks are contemporaneous surprises. It fits a model in which technological opportunities are learned about before they show up in measured productivity.

The evidence is strongest when TFP is adjusted for variable factor utilization. In that case, stock prices rise first, while TFP growth arrives only years later. Adding consumption, investment, and hours shows that the same shock also produces a temporary macroeconomic boom. Thus the paper provides an empirical foundation for expectation-driven business cycles based on news about future fundamentals.

9 Conclusion

Beaudry and Portier argue that expectations about future productivity are central to business cycles. The key empirical object is the near identity between two shocks: a stock-market innovation with no immediate TFP effect and the shock that drives long-run TFP. The paper interprets this identity as evidence that the stock market responds to news about future technological improvements.

The paper’s broader implication is methodological and substantive. Methodologically, stock prices should be used more directly in macroeconomic VAR analysis because they contain information about expectations. Substantively, business cycles may be driven not by contemporaneous productivity surprises but by agents’ advance recognition of future technological opportunities.

The final takeaway is that asset prices can help reveal the timing of information. If stock prices move first, real activity responds next, and measured TFP follows with a delay, then a boom can be fundamentally driven even though the fundamental productivity improvement has not yet appeared in the data.